

1 CORS Error Fix

Problem: Frontend (port 3000) cannot access backend (port 8020).

Solution: Enable CORS in Express backend.

```
// Install cors
npm install cors

const express = require('express');
const cors = require('cors');
const app = express();

// Allow frontend origin
app.use(cors({
  origin: 'http://localhost:3000',
  methods: ['GET', 'POST', 'PUT', 'DELETE'],
  credentials: true
}));

app.get('/', (req, res) => res.json({ msg: 'CORS fixed!' }));

app.listen(8020, () => console.log('Server running on port 8020'));
```

For testing only: allow all origins

```
app.use(cors());
```

2 Axios Request Formats

GET - Fetch all items

```
axios.get('http://localhost:8020/items')
  .then(res => console.log(res.data))
  .catch(err => console.error(err));
```

POST - Create a new item

```
axios.post('http://localhost:8020/items', { name: 'New Item' })
  .then(res => console.log(res.data))
  .catch(err => console.error(err));
```

PUT - Update an existing item

```
axios.put('http://localhost:8020/items/123', { name: 'Updated Item' })
  .then(res => console.log(res.data))
  .catch(err => console.error(err));
```

DELETE - Delete an item

```
axios.delete('http://localhost:8020/items/123')
  .then(res => console.log('Deleted'))
  .catch(err => console.error(err));
```

Tips: - Always wrap axios calls in `try/catch` if using `async/await`. - Ensure backend has proper CORS configuration. - `_id` is MongoDB default; replace if using a different key.

3 Steps to Link React Frontend with Express + Node + Mongoose Backend

1. **Backend Setup:**

2. Initialize Node project: `npm init -y`

3. Install dependencies: `express`, `mongoose`, `cors`, `dotenv`

4. Connect to MongoDB using Mongoose:

```
const mongoose = require('mongoose');
mongoose.connect(process.env.DBURL, { useNewUrlParser: true,
useUnifiedTopology: true })
  .then(() => console.log('MongoDB connected'))
  .catch(err => console.log('DB connection error', err));
```

5. **Enable CORS:** as shown above to allow frontend requests.

6. **Define API routes:** `/items` with GET, POST, PUT, DELETE.

7. **React Frontend:**

8. Use `axios` with the formats above to call backend API.

9. **Run:**

10. Backend: `node server.js` (port 8020)

11. Frontend: `npm start` (React dev server, port 3000)

12. **Check:** Frontend should fetch and manipulate data from backend seamlessly.

End of Guide